

## Interview Transcript with Dr. Siddharth Rao (Bengaluru, India) Senior Consultant, Cardiology, Manipal Hospitals

Interviewer: Dr. Rao, thank you for joining us. Could you start by sharing your background and what your role involved during the pandemic? Dr. Rao: Thank you. I'm Dr. Siddharth Rao, a cardiologist at Manipal Hospitals in Bengaluru. I've practiced for about 18 years, focusing on interventional cardiology and heart failure. During COVID, cardiologists were thrust into a different world—suddenly, the line between pulmonology and cardiology blurred.

We learned that COVID wasn't just a respiratory disease—it was a vascular one. Patients came with myocarditis, arrhythmias, and clotting disorders. I was part of our hospital's Cardio-COVID response unit, responsible for managing cardiac emergencies among COVID patients and developing protocols for thrombotic complications.

Interviewer: What was the first wave like for your department? Dr. Rao: The first wave was more of a psychological shock than a clinical one. Bengaluru's hospitals saw steady numbers but not overwhelming surges initially. The challenge was isolation and uncertainty. Elective cardiac procedures were halted. Cath labs were converted to negative-pressure emergency areas. We still had to treat heart attacks, of course—heart disease doesn't take a lockdown. The problem was that patients stopped coming. Many delayed treatment out of fear of infection. I remember one gentleman who stayed home with chest pain for two days; by the time he came, the damage was irreversible. That became a recurring tragedy—people dying of fear, not the virus itself.

Interviewer: When did cardiac complications start becoming evident? Dr. Rao: Around mid-2020, reports began emerging about myocarditis and microthrombosis. We started seeing COVID patients with elevated troponins, irregular rhythms, and heart failure without classic blockages. COVID inflamed the heart muscle in ways we hadn't seen before.

Later, during the Delta wave, clotting became the biggest issue—pulmonary embolisms, strokes, even young people with massive heart attacks. The virus damaged endothelium, making blood hypercoagulable. Our cardiology wards turned into hybrid COVID ICUs. We worked side by side with pulmonologists to manage overlapping complications.

Interviewer: How did your hospital adapt to handle those overlapping cases? Dr. Rao: It required a complete redesign of workflow. We created an integrated COVID Command model where cardiology, pulmonology, and internal medicine shared patient dashboards. Every severe COVID admission was screened for cardiac risk markers—troponin, D-dimer, CRP.

When D-dimer spiked, we'd intervene early with anticoagulants. We also developed a triage matrix—mild elevation, observe; moderate, start heparin; high, call cardiology. That proactive model reduced cardiac deaths significantly.

Tele-consultation also became vital. Many post-COVID cardiac patients were followed remotely to avoid hospital exposure. We learned that digital medicine, when structured, can save lives.

Interviewer: What was the situation like during the Delta wave in Bengaluru? Dr. Rao:

Chaotic, heartbreaking, and humbling. Bengaluru's healthcare system was stretched to the edge. Our 1000-bed hospital had 700 COVID patients at one point. Cardiology became a 24-hour service not just for COVID patients, but for recovered ones returning with heart issues.

Oxygen shortages hit even tertiary hospitals. I recall nights when cardiologists were literally managing oxygen triage, deciding whether to allocate high-flow oxygen to a cardiac patient or an intubated COVID case. These were impossible choices, made on the fly with no luxury of time or certainty.

Interviewer: Were there particular patterns in cardiac complications after recovery? Dr. Rao:

Yes, the so-called Long COVID heart. We saw lingering tachycardia, palpitations, and exercise intolerance. Some developed myocarditis weeks after recovery. The anxiety around chest pain became widespread every flutter felt like a heart attack to patients.

We created a Post-COVID Cardiac Clinic to screen patients for residual inflammation. About 20 percent showed measurable cardiac changes on echo or MRI, though most improved with rest and medication. It taught us that viral inflammation can leave invisible scars not just in lungs but in hearts too.

Interviewer: How did you manage staff morale and burnout through those months? Dr. Rao:

Burnout was universal. We were wearing PPE for 12 hours straight in 40-degree heat. At one point, four of our senior nurses were hospitalized themselves. We had to become each other's counselors.

The hospital arranged rotation breaks and psychological support, but the real coping mechanism was teamwork. Every successful recovery felt like a team medal. One ritual we kept ringing a small brass bell each time a COVID patient left the ICU. The sound carried down the corridor, and everyone paused, smiled, and breathed again.

Interviewer: How did the vaccination rollout affect cardiac admissions? Dr. Rao:

Dramatically. By mid-2021, once vaccination reached healthcare workers and older adults, severe cardiac-related COVID cases plummeted. The correlation was striking.

Later, when vaccine-related myocarditis became a talking point globally, we conducted an internal review of 60,000 vaccinated individuals. We found only two mild myocarditis cases both resolved fully. The benefits far outweighed the risks. We used those data to reassure hesitant patients. In public health, clear communication backed by data is half the cure.

Interviewer: How did misinformation impact your work? Dr. Rao:

It was exhausting. Patients came quoting YouTube videos about blood thinners causing bleeding deaths or vaccines blocking arteries. We had to spend hours correcting myths. The biggest damage misinformation

does is delay people wait too long to seek help.

We started community webinars titled Your Heart and COVID , where we simplified concepts explaining what clotting means, why oxygen helps, how vaccines actually reduce cardiac strain. When fear meets facts, healing becomes possible.

Interviewer: Could you share a particular story that stayed with you? Dr. Rao: I ll never forget a 42-year-old software engineer. He was recovering from mild COVID at home when he collapsed. We diagnosed a massive pulmonary embolism. He survived after 12 hours on ECMO the machine that oxygenates blood externally.

Weeks later, he visited with his wife and two kids, bringing a note that said, Thank you for giving me back the chance to be a father. It reminded us why we kept going despite exhaustion because behind every statistic is a family holding its breath.

Interviewer: Bengaluru has a strong tech ecosystem. Did data or AI tools play any role in your hospital s pandemic response? Dr. Rao: Yes, absolutely. We collaborated with a local health-tech startup to build predictive dashboards. Using anonymized EHR data, we could visualize oxygen usage, patient deterioration risk, and even predict which wards would fill next.

We also ran an AI pilot to detect cardiac involvement from chest CT scans it wasn t perfect but showed promise. COVID accelerated the fusion of medicine and data science by at least five years. I tell my students the next cardiologist will need to know code as well as catheterization.

Interviewer: Looking back, what do you think India s healthcare system learned from all this?

Dr. Rao: Three lessons stand out: 1. Integration beats specialization. Silos between departments waste time; multidisciplinary care saves lives. 2. Preparedness must include supply chains. Ventilators are useless without oxygen, and oxygen is useless without logistics. 3. Trust is infrastructure. When people trust doctors and data, compliance follows.

India s healthcare capacity expanded in months what would ve taken years oxygen plants, ICU networks, digital records. We must maintain that momentum, not slide back to complacency.

Interviewer: What about you personally what did the pandemic change in how you see medicine?

Dr. Rao: It stripped medicine down to its essence. During those nights when alarms were blaring and we were out of beds, titles didn t matter. Surgeons helped nurses, junior residents made triage calls, and administrators carried oxygen cylinders.

It also changed how I talk to patients. I now emphasize lifestyle, breathing, mental health things we used to delegate to other specialists. COVID taught us that the heart is not just an organ; it s a metaphor for resilience.

And on a human level, I learned the importance of gratitude. Every breath, every normal ECG,

feels like a blessing now.

Interviewer: Finally, if you had one message for young doctors entering medicine after this crisis, what would it be? Dr. Rao: Don't chase certainty chase understanding. Pandemics teach humility; medicine is not about having all the answers, but asking the right questions compassionately.

Also, protect your own wellbeing. You can't pour from an empty cup. The pandemic reminded us that healing others begins with caring for ourselves.

Interviewer: Thank you, Dr. Rao, for this powerful conversation. Dr. Rao: Thank you. These memories are heavy, but sharing them ensures we never forget what they taught us.

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